

## Packet Tracer - Use a TFTP Server to Upgrade a Cisco IOS Image

**Instructor Note:** Red font color or gray highlights indicate text that appears in the instructor copy only.

### Addressing Table

| Device      | Interface | IP Address    | Subnet Mask   | Default Gateway |
|-------------|-----------|---------------|---------------|-----------------|
| R1          | G0/0/0    | 192.168.2.1   | 255.255.255.0 | N/A             |
| R2          | G0/0      | 192.168.2.2   | 255.255.255.0 | N/A             |
| S1          | VLAN 1    | 192.168.2.3   | 255.255.255.0 | 192.168.2.1     |
| TFTP Server | NIC       | 192.168.2.254 | 255.255.255.0 | 192.168.2.1     |

### Objectives

**Part 1: Upgrade an IOS Image on a Cisco Device**

**Part 2: Backup an IOS Image on a TFTP Server**

### Scenario

A TFTP server can help manage the storage of IOS images and revisions to IOS images. For any network, it is good practice to keep a backup copy of the Cisco IOS Software image in case the system image in the router becomes corrupted or accidentally erased. A TFTP server can also be used to store new upgrades to the IOS and then deployed throughout the network where it is needed. In this activity, you will upgrade the IOS images on Cisco devices by using a TFTP server. You will also backup an IOS image with the use of a TFTP server.

### Instructions

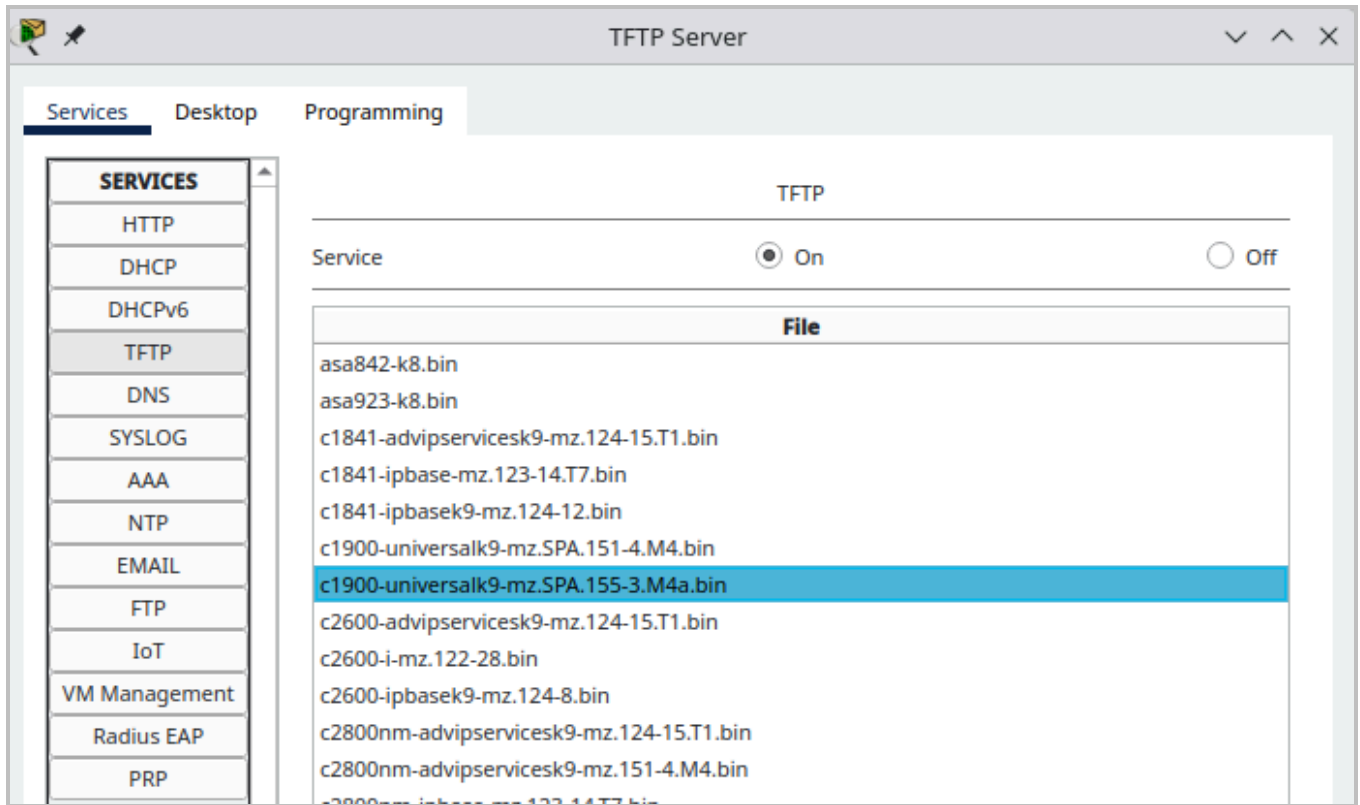
#### Part 1: Upgrade an IOS Image on a Cisco Device

##### Step 1: Upgrade an IOS image on a router.

- Access the TFTP server and enable the TFTP service.
- Note the IOS image files that are available on the TFTP server.

Which IOS images stored on the server are compatible with a 1941 router?

**c1900-universalk9-mz.SPA.151-4.M4.bin and c1900-universalk9-mz.SPA.155-3.M4a.bin**



- c. From **R2**, issue the **show flash:** command and record the available flash memory.

```
R2#
R2#show flash:

System flash directory:
File Length Name/status
  3 33591768 c1900-universalk9-mz.SPA.151-4.M4.bin
  5 33591768 c1900-universalk9-mz.SPA.155-3.M4a.bin
  2 28282 sigdef-category.xml
  1 227537 sigdef-default.xml
[67439355 bytes used, 188304645 available, 255744000 total]
249856K bytes of processor board System flash (Read/Write)
```

**221896413**

- d. Copy the CISCO1941/K9 IOS version 15.5 image for the 1941 router from the TFTP Server to R2.

**Note:** In an actual network, if there is more than one interface active on the router, you may need to enter the **ip tftp source interface** command to specify which interface should be used to contact the TFTP server. This command is not supported in PT 7.2 and older versions and is not necessary to complete this activity.

R2# **copy tftp: flash:**

Address or name of remote host []? **192.168.2.254**

Source filename []? **c1900-universalk9-mz.SPA.155-3.M4a.bin**

Destination filename [c1900-universalk9-mz.SPA.155-3.M4a.bin]?

Accessing tftp://192.168.2.254/c1900-universalk9-mz.SPA.155-3.M4a.bin....

Loading c1900-universalk9-mz.SPA.155-3.M4a.bin from 192.168.2.254:

```
!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!
!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!
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[OK - 33591768 bytes]

R2>enable
R2#copy tftp: flash:
Address or name of remote host []? 192.168.2.254
Source filename []? c1900-universalk9-mz.SPA.155-3.M4a.bin
Destination filename [c1900-universalk9-mz.SPA.155-3.M4a.bin]?

Accessing tftp://192.168.2.254/c1900-universalk9-mz.SPA.155-3.M4a.bin...
Loading c1900-universalk9-mz.SPA.155-3.M4a.bin from 192.168.2.254:
!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!
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[OK - 33591768 bytes]

33591768 bytes copied in 3.476 secs (1014671 bytes/sec)
R2#
```

33591768 bytes copied in 4.099 secs (860453 bytes/sec)

- e. Verify that the IOS image has been copied to flash.

How many IOS images are located in flash?

**2**

- f. Use the **boot system** command to load the version 15.5 IPBase image on the next reload.

R2(config)# **boot system flash c1900-universalk9-mz.SPA.155-3.M4a.bin**

```
R2#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
R2(config)#boot system flash c1900-universalk9-mz.SPA.155-3.M4a.bin
R2(config)#exit
R2#
%SYS-5-CONFIG_I: Configured from console by console

R2#copy running-config startup-config
Destination filename [startup-config]?
Building configuration...
[OK]
R2#
```

- g. Save the configuration and reload **R2**.

```
R2#reload
Proceed with reload? [confirm]
System Bootstrap, Version 15.1(4)M4, RELEASE SOFTWARE (fc1)
Technical Support: http://www.cisco.com/techsupport
Copyright (c) 2010 by cisco Systems, Inc.
Total memory size = 512 MB - On-board = 512 MB, DIMM0 = 0 MB
CISCO1941/K9 platform with 524288 Kbytes of main memory
Main memory is configured to 64/-1(On-board/DIMM0) bit mode with ECC disabled

Readonly ROMMON initialized

program load complete, entry point: 0x80803000, size: 0x1b340
program load complete, entry point: 0x80803000, size: 0x1b340

IOS Image Load Test
-----
Digitally Signed Release Software
program load complete, entry point: 0x81000000, size: 0x2bb1c58
Self decompressing the image :
##### [OK]
Smart Init is enabled
smart init is sizing iomem
      TYPE      MEMORY_REQ
Onboard devices &
  buffer pools    0x01E8F000
-----
      TOTAL:      0x01E8F000
Rounded IOMEM up to: 32Mb.
Using 6 percent iomem. [32Mb/512Mb]
```

- h.
- i. Use the **show version** command to verify the upgraded IOS image is loaded after **R2** reboots.

R2# **show version**

License Info:

License UDI:

| Device# | PID          | SN          |
|---------|--------------|-------------|
| *0      | CISCO1941/K9 | FTX15242KPQ |

Suite License Information for Module:'c1900'

| Suite                                     | Suite Current | Type | Suite Next reboot |
|-------------------------------------------|---------------|------|-------------------|
| FoundationSuiteK9<br>securityk9<br>datak9 | None          | None | None              |

Technology Package License Information for Module:'c1900'

| Technology | Technology-package<br>Current | Type      | Technology-package<br>Next reboot |
|------------|-------------------------------|-----------|-----------------------------------|
| ipbase     | ipbasek9                      | Permanent | ipbasek9                          |
| security   | disable                       | None      | None                              |
| data       | disable                       | None      | None                              |

Configuration register is 0x2102

R2>

### Step 2: Upgrade an IOS image on a switch.

- Access the TFTP server and copy the **c2960-lanbasek9-mz.150-2.SE4.bin** image to S1.

S1# copy tftp: flash:

```
S1>copy tftp: flash:
^
% Invalid input detected at '^' marker.

S1>enable
S1#copy tftp: flash:
Address or name of remote host []? 192.168.2.254
Source filename []? c2960-lanbasek9-mz.150-2.SE4.bin
Destination filename [c2960-lanbasek9-mz.150-2.SE4.bin]?

Accessing tftp://192.168.2.254/c2960-lanbasek9-mz.150-2.SE4.bin....
Loading c2960-lanbasek9-mz.150-2.SE4.bin from 192.168.2.254:
!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!
[OK - 4670455 bytes]

4670455 bytes copied in 3.043 secs (123393 bytes/sec)
```

- Use the **boot system** command to configure the switch to load the new IOS image on boot.
- Reload S1 and verify the new image has been loaded into memory.
- Close the TFTP configuration window if it is still open.

```
S1#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
S1(config)#boot system flash:c2960-lanbasek9-mz.150-2.SE4.bin
S1(config)#exit
S1#
%SYS-5-CONFIG_I: Configured from console by console

S1#copy running-config startup-config
Destination filename [startup-config]?
Building configuration...
[OK]
S1#reload
Proceed with reload? [confirm]
C2960 Boot Loader (C2960-HBOOT-M) Version 12.2(25r)FX, RELEASE SOFTWARE (fc4)
Cisco WS-C2960-24TT (RC32300) processor (revision C0) with 21039K bytes of memory.
2960-24TT starting...
Base ethernet MAC Address: 0060.70E3.A7AB
Xmodem file system is available.
Initializing Flash...
flashfs[0]: 3 files, 0 directories
flashfs[0]: 0 orphaned files, 0 orphaned directories
flashfs[0]: Total bytes: 64016384
flashfs[0]: Bytes used: 9342038
flashfs[0]: Bytes available: 54674346
flashfs[0]: flashfs fsck took 1 seconds.
...done Initializing Flash.

Boot Sector Filesystem (bs:) installed, fsid: 3
Parameter Block Filesystem (pb:) installed, fsid: 4

Loading "flash:/c2960-lanbasek9-mz.150-2.SE4.bin"...
##### [OK]

The password-recovery mechanism is enabled.
64K bytes of flash-simulated non-volatile configuration memory.
Base ethernet MAC Address      : 00:60:70:E3:A7:AB
Motherboard assembly number    : 73-10390-03
Power supply part number      : 341-0097-02
Motherboard serial number      : FOC10093R12
Power supply serial number     : AZS1007032H
Model revision number          : B0
Motherboard revision number    : B0
Model number                   : WS-C2960-24TT-L
System serial number           : FOC1010X104
Top Assembly Part Number       : 800-27221-02
Top Assembly Revision Number   : A0
Version ID                     : V02
CLEI Code Number               : COM3L00BRA
Hardware Board Revision Number : 0x01

Switch Ports Model          SW Version  SW Image
-----
*   1 26    WS-C2960-24TT-L    15.0(2)SE4    C2960-LANBASEK9-M

Cisco IOS Software, C2960 Software (C2960-LANBASEK9-M), Version 15.0(2)SE4, RELEASE SOFTWARE (fc1)
Technical Support: http://www.cisco.com/techsupport
Copyright (c) 1986-2013 by Cisco Systems, Inc.
Compiled Wed 26-Jun-13 02:49 by mnnguyen
```

## Part 2: Backup an IOS Image to a TFTP Server

- a. On **R1**, display the contents of flash and record the IOS image.

**isr4300-universalk9.03.16.05.S.155-3.S5-ext.SPA.bin**

```
R1# show flash:
```

```
R1>enable
R1#show flash:

System flash directory:
File      Length      Name/status
  3    486899872 2isr4300-universalk9.03.16.05.S.155-3.S5-ext.SPA.bin
  2    28282      sigdef-category.xml
  1    227537      sigdef-default.xml
[487155691 bytes used, 2761893909 available, 3249049600 total]
3.17338e+06K bytes of processor board System flash (Read/Write)

R1#
```

- b. Use the **copy** command to back up the IOS image in flash memory on **R1** to a TFTP server. **Note:** The isr4300 image is considerably larger than the c1900 image. It will take longer to transmit it to the TFTP server.

```
R1# copy flash: tftp:
```

Source filename []? **isr4300-universalk9.03.16.05.S.155-3.S5-ext.SPA.bin**

```
Address or name of remote host []? 192.168.2.254
```

Destination filename [isr4300-universalk9.03.16.05.S.155-3.S5-ext.SPA.bin]?

[illegible]

```
486899872 bytes copied in 18.815 secs (83367 bytes/sec)
```

```
R1#copy flash: tftp:
Source filename []? isr4300-universalk9.03.16.05.S.155-3.S5-ext.SPA.bin
Address or name of remote host []? 192.168.2.254
Destination filename [isr4300-universalk9.03.16.05.S.155-3.S5-ext.SPA.bin]?

Writing isr4300-universalk9.03.16.05.S.155-3.S5-
ext.SPA.bin.....
```

- c. Access the TFTP server and verify that the IOS image has been copied to the TFTP server.

**Note:** You may have to start and stop the TFTP service on the server so the file appears in the file listing.

## Packet Tracer - Use a TFTP Server to Upgrade a Cisco IOS Image

